AI BUILD CHALLENGE — BENGALURU

INDUSTRY PROBLEM STATEMENT

Presented by Masters' Union Ventures | Powered by Hackingly

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PS-001 · QUALCOMM EDGE AI TRACK

Offline, Privacy-First Document Intelligence (On-Device RAG)

INDUSTRY PARTNER: **QUALCOMM**

THE CHALLENGE

People and organizations hold large amounts of sensitive documents today — medical records, legal contracts, personal notes, internal company files — that they'd like to query and summarize with AI. Sending this data to cloud LLM APIs creates real privacy and compliance risk, and many real-world settings such as fieldwork, secure facilities, and low-connectivity areas don't have reliable internet at all. The challenge is to let a user query and reason over their own local documents entirely on-device, with zero data ever leaving the phone. Build a fully offline Retrieval-Augmented Generation (RAG) application on Snapdragon that lets a user import personal or organizational documents — PDFs, notes, scanned pages — and ask natural-language questions against them, with the entire embedding, retrieval, and generation pipeline running on-device.

YOUR MISSION

Build an AI-powered solution that addresses this problem and demonstrates a clear, practical use case.

- Let users import personal or organizational documents and query them in natural language, fully offline
- Run the entire embedding, retrieval, and generation pipeline on-device on Snapdragon
- Produce grounded, cited answers rather than hallucinated ones

WHO IS THE USER?

Primary User: Individuals and organizations handling sensitive documents (medical, legal, personal, internal company files)

Context: They need to query and summarize these documents with AI but cannot send them to the cloud, due to privacy or compliance risk, or because they're working in fieldwork, secure, or low-connectivity settings.

WHAT COULD YOUR SOLUTION DO?

- Index and embed documents (PDFs, notes, scanned pages) fully on-device
- Keep retrieval fast and accurate as the local document set grows
- Generate grounded, cited answers using Genie/GenieX or llama.cpp
- Handle scanned/image-based documents via on-device OCR and reason across multiple documents

These are starting points, not limitations. Teams are encouraged to think beyond the suggested directions.

KEY CONSIDERATIONS

- Zero data may leave the device — no cloud calls at any point
- Retrieval speed and accuracy must hold up as the document set grows
- Must operate within realistic Snapdragon memory and compute constraints
- Generation should be grounded and cited, not hallucinated

Resources / APIs / Data: Genie/GenieX, llama.cpp, Qualcomm AI Hub

BUILD EXPECTATIONS

IDEA → BUILD → ITERATE → TEST → SHIP

You can start building as soon as the problem statement is shared. Come to the venue with approximately 80% of your solution ready. The final product must be built, tested and completed at the venue during the AI Build Challenge.

FINAL EVALUATION: 100% WEIGHTAGE | WORKING PROTOTYPE REQUIRED

Problem Understanding & Relevance • Solution Quality & Problem-Solution Fit • Execution & Working Prototype • Innovation / Use of AI • Feasibility & Potential Impact

Don't just talk about AI. Build with it.